What k-means guarantees, and what it does not

BIS 527: Introduction to Health Data Science Lecture 11, October 8, 2026

Notation

Every symbol and term used in these notes

Points	
n, i	The number of points, and a point. Points are numbered $i = 0, 1, \ldots, n-1$.
p, j	The number of coordinates per point, and a coordinate, numbered $j = 0, \ldots, p-1$.
x_i, $x_i[j]$	Point i, $x_i = (x_i[0], \ldots, x_i[p-1])$, and its coordinate j.
$\|\|x - y\|\|^2$	The squared distance $\sum_{j=0}^{p-1}(x[j] - y[j])^2$ between points x and y.
$\|S\|$, $\bar{x}_S$	The number of points in a set S, and, when S is nonempty, their mean: $\bar{x}_S[j] = \frac{1}{\|S\|}\sum_{i \in S} x_i[j]$. The bar on $\bar{x}_S$ marks a mean, while the bars on $\bar{C}$, $\bar{\mu}$ and $\bar{m}$ mark lists.
Clusters and centers	
K, k	The number of clusters, and a cluster number, $k = 1, \ldots, K$.
$\bar{C}$, C_k	A partition $\bar{C} = (C_1, \ldots, C_K)$: sets of points that do not overlap and together contain every point. C_k is cluster k, and it may be empty.
$\bar{\mu}$, μ_k	Centers (also called focal points): $\bar{\mu} = (\mu_1, \ldots, \mu_K)$, points with p coordinates.
nearest center	A center μ_k with the smallest $\|\|x_i - \mu_k\|\|^2$. A point can have several nearest centers.
Cost	
$W(\bar{C}, \bar{\mu})$	The cost of the partition $\bar{C}$ with the centers $\bar{\mu}$: $\sum_{k=1}^{K}\sum_{i \in C_k}\|\|x_i - \mu_k\|\|^2$ (Definition 1).
W^*	The smallest cost over all partitions and all centers (Definition 2).
$\bar{m}$, m_k	The means of the clusters: $m_k = \bar{x}_{C_k}$ for every nonempty C_k, and $m_k = \mu_k$ for every empty one (Theorem 1).
$\bar{C}^{\text{nc}}$	A nearest-center partition for $\bar{\mu}$: every point is in the cluster of one of its nearest centers (Theorem 2).
$\bar{C}^*$, $\bar{\mu}^*$	A best pair: a partition and centers with $W(\bar{C}^*, \bar{\mu}^*) = W^*$ (Definition 2).
$\mathcal{R}$	The set of all stopping points of k-means (Definition 3).
G, L	In Theorem 6, any positive number, and the width of a rectangle with $L^2 > G$.
$F(\bar{\mu})$	The loss of the centers: $\min_{\bar{C}} W(\bar{C}, \bar{\mu}) = \sum_i \min_k \|\|x_i - \mu_k\|\|^2$ (Section 8).
The algorithm	
round	One partition step, followed by one learning step unless the algorithm stops.
tie rule	In the partition step, a point stays in its current cluster when that cluster's center is among its nearest.
$\bar{C}'$, $\bar{\mu}'$	The partition after a partition step, and the centers after a learning step (Theorem 3).
$\bar{C}^{(r)}$, $\bar{\mu}^{(r)}$	The partition after partition step r, and the centers after learning step r. $\bar{\mu}^{(0)}$ are the starting centers. Rounds are numbered from 1.
W_r	$W(\bar{C}^{(r)}, \bar{\mu}^{(r)})$, the cost after learning step r.
Shape of a function (Section 8)	
g	Any function that takes a list of K centers and returns a number, such as F.
$\bar{a}$, $\bar{b}$, $\bar{c}$, $\bar{\nu}$	Lists of K centers, such as $\bar{\nu} = (\nu_1, \ldots, \nu_K)$.
$(1-\lambda)\bar{a} + \lambda\bar{b}$	For $0 \le \lambda \le 1$, the list with centers $(1-\lambda)a_k + \lambda b_k$, on the straight segment from $\bar{a}$ to $\bar{b}$.
local, global minimum	Definition 4. Convex function: Definition 5.

1 Recap

Points, clusters, centers, and the algorithm

Points, clusters and centers. The data are n points $x_0, \ldots, x_{n-1}$, each with p coordinates. A *partition* $\bar{C} = (C_1, \ldots, C_K)$ splits them into K clusters: sets of points that do not overlap and together contain every point. A cluster may be empty. Each cluster k has a *center* μ_k, and we collect the centers in $\bar{\mu} = (\mu_1, \ldots, \mu_K)$. In Lecture 10, the points were households, and van k parked at μ_k and served the households of C_k.

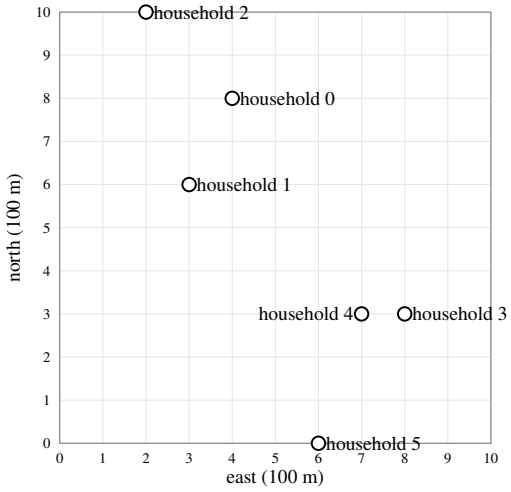
The algorithm. k-means starts from K starting centers and repeats rounds.

1. **Partition step** (changes $\bar{C}$). Every point goes to the cluster of a nearest center. *Tie rule:* a point stays in its current cluster when that cluster's center is among its nearest. Otherwise it goes to the nearest center with the smallest number.

2. **Learning step** (changes $\bar{\mu}$). Every center moves to the mean of its cluster: μ_k becomes $\bar{x}_{C_k}$ for every nonempty C_k. The center of an empty cluster stays where it is.

Stopping rule. If a partition step moves no point, the algorithm stops and returns $\bar{C}$ and $\bar{\mu}$. Placing a point for the first time counts as a move, so the first round always has a learning step.

The running example. Survey A has six households, with coordinates east and north in units of 100 m. With $K = 2$ and the starting centers at households 0 and 1, k-means runs as follows.



Step	Partition, or centers
start	$\mu_1 = (4, 8)$, $\mu_2 = (3, 6)$
partition 1	$C_1 = \{0, 2\}$, $C_2 = \{1, 3, 4, 5\}$
learning 1	$\mu_1 = (3, 9)$, $\mu_2 = (6, 3)$
partition 2	$C_1 = \{0, 1, 2\}$, $C_2 = \{3, 4, 5\}$
learning 2	$\mu_1 = (3, 8)$, $\mu_2 = (7, 2)$
partition 3	no point moves: stop

2 The cost

What k-means makes small

Definition 1 (the cost, also called the within-cluster sum of squares)

For a partition $\bar{C} = (C_1, \ldots, C_K)$ and centers $\bar{\mu} = (\mu_1, \ldots, \mu_K)$, the *cost* is

$$W(\bar{C}, \bar{\mu}) := \sum_{k=1}^{K} \sum_{i \in C_k} \|x_i - \mu_k\|^2.$$

An empty cluster contributes nothing, whatever its center is.

For the vans, $W(\bar{C}, \bar{\mu})$ is the total squared distance from the households to the vans they use. Squaring weighs long trips heavily: a household twice as far from its van pays four times as much. Because every point lies in exactly one cluster, the same total can be added up point by point: each point contributes the squared distance to the center of its own cluster.

Definition 2 (the k-means problem)

Given the points and K, find a partition $\bar{C}$ and centers $\bar{\mu}$ with the smallest cost $W(\bar{C}, \bar{\mu})$. We write W^* for that smallest cost, and call a pair $(\bar{C}^*, \bar{\mu}^*)$ with $W(\bar{C}^*, \bar{\mu}^*) = W^*$ a *best pair*. Theorem 5 shows that a best pair exists.

The k-means problem has two unknowns, $\bar{C}$ and $\bar{\mu}$, and k-means handles them one at a time: the partition step changes only $\bar{C}$, and the learning step changes only $\bar{\mu}$. Theorems 1 and 2 show that each step finds the best value of its own unknown, given the other. A method that improves one unknown at a time in this way is called *alternating minimization*.

The running example. The cost after each step of the run in Section 1 never goes up.

After	partition 1	learning 1	partition 2	learning 2	partition 3
W	112	36	27	18	18

For example, after learning step 2 the centers are $\mu_1 = (3, 8)$ and $\mu_2 = (7, 2)$. The households of C_1 pay $1 + 4 + 5$ and those of C_2 pay $2 + 1 + 5$, so $W = 18$.

3 Theorem 1

Given the partition, the means are the best centers

Theorem 1 (given the partition, the means are best)

Let $\bar{C}$ be a partition, let $\bar{\mu}$ be any centers, and let $\bar{m}$ be the means of the clusters: $m_k = \bar{x}_{C_k}$ for every nonempty cluster C_k, and $m_k = \mu_k$ for every empty one. Then

$$W(\bar{C}, \bar{\mu}) = W(\bar{C}, \bar{m}) + \sum_{k:\, C_k \neq \{\}} |C_k| \, \|m_k - \mu_k\|^2.$$

So $W(\bar{C}, \bar{m}) \leq W(\bar{C}, \bar{\mu})$, with equality only when $\mu_k = m_k$ for every nonempty cluster.

Proof

Step 1: one set, deviations add to zero. Let S be a nonempty set of points with mean $m = \bar{x}_S$. For each coordinate j, the definition of the mean gives $|S|\, m[j] = \sum_{i \in S} x_i[j]$, so $\sum_{i \in S} \big(x_i[j] - m[j]\big) = 0$.
Step 2: one set, one coordinate. Let c be any point. Write $x_i[j] - c[j] = \big(x_i[j] - m[j]\big) + \big(m[j] - c[j]\big)$ and square:

$$\big(x_i[j] - c[j]\big)^2 = \big(x_i[j] - m[j]\big)^2 + 2\big(x_i[j] - m[j]\big)\big(m[j] - c[j]\big) + \big(m[j] - c[j]\big)^2.$$

Add over $i \in S$. The factor $m[j] - c[j]$ does not depend on i, so the middle terms add to $2\big(m[j] - c[j]\big) \sum_{i \in S} \big(x_i[j] - m[j]\big)$, which is 0 by Step 1. The last term is the same for every i, so it adds to $|S|\big(m[j] - c[j]\big)^2$. Hence

$$\sum_{i \in S} \big(x_i[j] - c[j]\big)^2 = \sum_{i \in S} \big(x_i[j] - m[j]\big)^2 + |S|\big(m[j] - c[j]\big)^2.$$

Step 3: one set, all coordinates. Adding Step 2 over $j = 0, \ldots, p-1$ gives

$$\sum_{i \in S} \|x_i - c\|^2 = \sum_{i \in S} \|x_i - m\|^2 + |S| \, \|m - c\|^2.$$

Step 4: all clusters. By Definition 1, $W(\bar{C}, \bar{\mu})$ is the sum over k of $\sum_{i \in C_k} \|x_i - \mu_k\|^2$. An empty cluster contributes nothing to $W(\bar{C}, \bar{\mu})$ or to $W(\bar{C}, \bar{m})$. For a nonempty cluster, Step 3 with $S = C_k$ and $c = \mu_k$ gives

$$\sum_{i \in C_k} \|x_i - \mu_k\|^2 = \sum_{i \in C_k} \|x_i - m_k\|^2 + |C_k| \, \|m_k - \mu_k\|^2,$$

and the first sum on the right is the contribution of cluster k to $W(\bar{C}, \bar{m})$. Adding over the nonempty clusters gives the equation of the theorem.
Step 5: the best centers. Every term $|C_k| \, \|m_k - \mu_k\|^2$ is at least 0, and it is 0 only when $\mu_k = m_k$. So $W(\bar{C}, \bar{m}) \leq W(\bar{C}, \bar{\mu})$, with equality only when every nonempty cluster's center is its mean.

Example. After partition step 1 of the running example, the cost is 112 at the centers $(4, 8)$, $(3, 6)$ and 36 at the means $(3, 9)$, $(6, 3)$. The extra terms are $2 \cdot 2 = 4$ and $4 \cdot 18 = 72$, and indeed $112 = 36 + 4 + 72$.

Implications. Theorem 1 is the reason the learning step uses means. It depends on the *squared* distance: with the plain distance, the best center is in general not the mean (for one coordinate it is a median).

4 Theorem 2

Given the centers, the nearest-center partition is the best

Theorem 2 (given the centers, the nearest-center partition is optimal)

Let $\bar{\mu}$ be centers, and let $\bar{C}^{\text{nc}}$ be a partition in which every point is in the cluster of one of its nearest centers. Then for every partition $\bar{C}$,

$$W(\bar{C}, \bar{\mu}) \;\geq\; W(\bar{C}^{\text{nc}}, \bar{\mu}) = \sum_{i=0}^{n-1} \min_{k=1,\ldots,K} \|x_i - \mu_k\|^2,$$

with equality only when every point of $\bar{C}$ is in the cluster of one of its nearest centers.

Proof

Step 1: the cost of $\bar{C}^{\text{nc}}$, point by point. Every point i lies in exactly one cluster of $\bar{C}^{\text{nc}}$, whose center is a nearest center of x_i, so point i contributes $\min_k \|x_i - \mu_k\|^2$. Adding over the points gives the equality on the right.

Step 2: one point. Let $\bar{C}$ be any partition, and let point i lie in its cluster C_l. The number $\|x_i - \mu_l\|^2$ is one of the K numbers $\|x_i - \mu_1\|^2, \ldots, \|x_i - \mu_K\|^2$, and the smallest of K numbers is at most each of them, so $\|x_i - \mu_l\|^2 \geq \min_k \|x_i - \mu_k\|^2$.

Step 3: all points. Adding Step 2 over $i = 0, \ldots, n-1$ adds up $W(\bar{C}, \bar{\mu})$ point by point on the left, so $W(\bar{C}, \bar{\mu}) \geq W(\bar{C}^{\text{nc}}, \bar{\mu})$.

Step 4: equality. The sum of the inequalities of Step 2 is an equality only when every one of them is, that is, when every point of $\bar{C}$ is in the cluster of one of its nearest centers.

Example. At partition step 2 of the running example, the centers are $\mu_1 = (3, 9)$ and $\mu_2 = (6, 3)$, and the partition $C_1 = \{0, 2\}$, $C_2 = \{1, 3, 4, 5\}$ costs 36. Household 1, at $(3, 6)$, is at squared distance 9 from $(3, 9)$ and 18 from $(6, 3)$, so its nearest center is μ_1. Moving it to C_1 gives the nearest-center partition $C_1 = \{0, 1, 2\}$, $C_2 = \{3, 4, 5\}$, its term drops from 18 to 9, and the cost from 36 to 27.

Implications. First, the k-means problem can be written with the centers alone. For any centers the best partition is a nearest-center partition, so W^* is the smallest value of $\sum_i \min_k \|x_i - \mu_k\|^2$ over all centers $\bar{\mu}$. This is how the problem was posed for the vans: choose where to park, and every household goes to the nearest van. Section 8 studies this function of the centers. Second, the nearest-center partition does not depend on how many points each cluster receives. With vans at east coordinates 0 and 100 and households at 1, 2 and 3, the van at 100 serves no household.

5 Theorem 3

Every step keeps the cost from going up

Theorem 3 (the cost goes down)

(a) If a partition step, with centers $\bar{\mu}$, turns the partition $\bar{C}$ into $\bar{C}'$, then $W(\bar{C}', \bar{\mu}) \leq W(\bar{C}, \bar{\mu})$, and the inequality is strict when some point moved.

(b) If a learning step, with partition $\bar{C}$, turns the centers $\bar{\mu}$ into $\bar{\mu}'$, then $W(\bar{C}, \bar{\mu}') \leq W(\bar{C}, \bar{\mu})$.

(c) Hence, for every round $r \geq 1$ after which another round moves a point, $W_{r+1} < W_r$.

Proof

Part (a). The centers do not change in a partition step. Take a point i, in cluster C_k before the step and in cluster C'_l after it. If $l = k$, its contribution $\|x_i - \mu_k\|^2$ is the same before and after. If $l \neq k$, the tie rule says the point moved only because μ_k is not among its nearest centers, while μ_l is a nearest center. So $\|x_i - \mu_l\|^2 = \min_{k'} \|x_i - \mu_{k'}\|^2 < \|x_i - \mu_k\|^2$. Adding over all points gives $W(\bar{C}', \bar{\mu}) \leq W(\bar{C}, \bar{\mu})$, strictly when at least one point moved.

Part (b). The partition does not change in a learning step, and the new centers are $\mu'_k = \bar{x}_{C_k}$ for every nonempty cluster and $\mu'_k = \mu_k$ for every empty one. These are the means $\bar{m}$ of Theorem 1, so $W(\bar{C}, \bar{\mu}') = W(\bar{C}, \bar{m}) \leq W(\bar{C}, \bar{\mu})$.

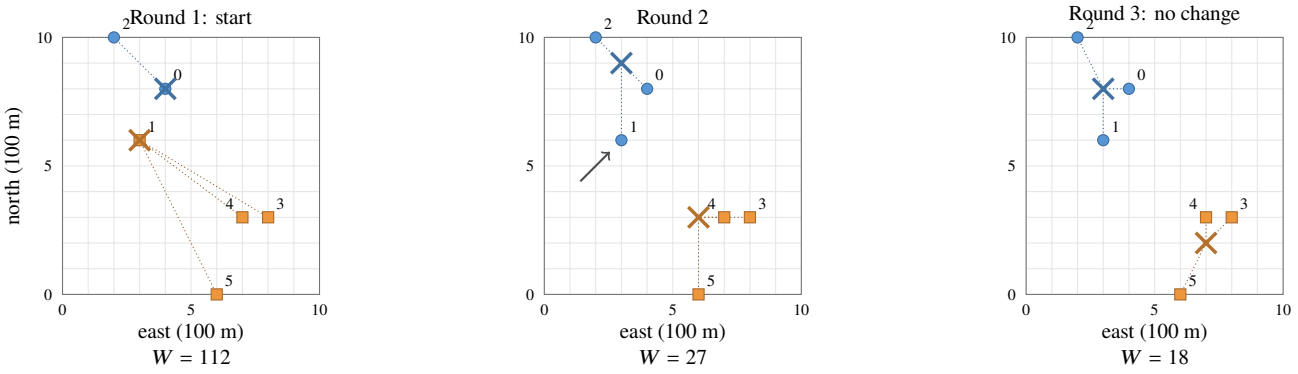
Part (c). Suppose the partition step of round $r + 1$ moves a point. It turns $\bar{C}^{(r)}$ into $\bar{C}^{(r+1)}$ with the centers $\bar{\mu}^{(r)}$, so part (a) gives

$$W\big(\bar{C}^{(r+1)}, \bar{\mu}^{(r)}\big) < W\big(\bar{C}^{(r)}, \bar{\mu}^{(r)}\big) = W_r.$$

Its learning step then turns $\bar{\mu}^{(r)}$ into $\bar{\mu}^{(r+1)}$ with the partition $\bar{C}^{(r+1)}$, and part (b) gives

$$W_{r+1} = W\big(\bar{C}^{(r+1)}, \bar{\mu}^{(r+1)}\big) \leq W\big(\bar{C}^{(r+1)}, \bar{\mu}^{(r)}\big).$$

Together, $W_{r+1} < W_r$.



The three partition steps of the running example. Crosses are the centers each step uses.

Example. In the running example, part (b) is the drop from 112 to 36 and from 27 to 18, part (a) is the drop from 36 to 27 when household 1 moved, and part (c) is $W_2 = 18 < W_1 = 36$.

Implication. Part (a) needs the tie rule. Without it, a point could move between two equally near centers without lowering the cost, and the inequality in part (a) would not be strict.

6 Theorem 4

k-means stops

Theorem 4 (eventually it stops)

For any points, any K and any starting centers, k-means stops after at most K^n learning steps and at most $K^n + 1$ partition steps.

Proof

Step 1: names. Let $\bar{C}^{(r)}$ be the partition after partition step r, let $\bar{\mu}^{(r)}$ be the centers after learning step r, and let $\bar{\mu}^{(0)}$ be the starting centers. For every round r that has a learning step, let $W_r = W(\bar{C}^{(r)}, \bar{\mu}^{(r)})$.

Step 2: W_r falls from round to round. A learning step happens in round $r + 1$ only when partition step $r + 1$ moves a point. So, by Theorem 3(c), $W_{r+1} < W_r$ for every pair of rounds $r \geq 1$ and $r + 1$ that both have a learning step.

Step 3: W_r depends only on the partition. After learning step r, the center of every nonempty cluster of $\bar{C}^{(r)} = (C_1, \ldots, C_K)$ is its mean. By Definition 1,

$$W_r = \sum_{k:\, C_k \neq \{\}} \sum_{i \in C_k} \|x_i - \bar{x}_{C_k}\|^2,$$

because an empty cluster contributes nothing, whatever center it kept. The right side is computed from $\bar{C}^{(r)}$ alone. So if $\bar{C}^{(s)} = \bar{C}^{(t)}$, then $W_s = W_t$.

Step 4: no partition repeats. Suppose the algorithm has completed R learning steps. By Step 2, $W_1 > W_2 > \cdots > W_R$. Any two of these numbers differ, so by Step 3 any two of $\bar{C}^{(1)}, \ldots, \bar{C}^{(R)}$ differ.

Step 5: counting. There are only K^n partitions into K numbered clusters, since each of the n points goes to one of K clusters. So by Step 4, $R \leq K^n$ whenever R learning steps have been completed, and the algorithm cannot complete $K^n + 1$ of them. Partition step $r + 1$ runs only after learning step r, so if R learning steps run, at most $R + 1$ partition steps run, and partition step $R + 1$ stops the algorithm, since otherwise learning step $R + 1$ would follow.

Implications. Compare the measure argument of Lecture 2. There, a recursive procedure stops because a nonnegative whole number falls at every call, and such a number cannot fall forever. Here, W_r falls at every round, but it is not a whole number in general, since the means can be fractions. What replaces "whole number" is Step 3: W_r is fixed by the partition, so it can take only as many values as there are partitions.

Common mistake. "The cost goes down every round, so k-means must stop." A falling sequence of numbers can go on forever, as $1, \frac{1}{2}, \frac{1}{4}, \ldots$ does. Steps 3 to 5 finish the proof: the finite number of partitions forces the stop.

The bound K^n shows only that k-means stops. For 100 points and $K = 2$ it allows more than 10^{30} rounds, while survey A needed two learning steps.

7 Stopping points

Where k-means can stop, and how good that is

Convergence. By Theorems 3 and 4, every run of k-means stops after finitely many rounds, and the cost falls at every round until then. Where a run stops depends on the starting centers, so we collect all the places where a run can stop.

Definition 3 (stopping points)

Fix the points and K. A pair $(\bar{C}, \bar{\mu})$ is a *stopping point* of k-means if k-means, started at some centers, stops and returns $\bar{C}$ and $\bar{\mu}$. We write $\mathcal{R}$ for the set of all stopping points.

Every stopping point $(\bar{C}, \bar{\mu})$ is a *fixed point* of k-means. The last partition step moved no point, so by the tie rule every point is in the cluster of one of its nearest centers, and the last learning step put the center of every nonempty cluster at its mean. So another round would change neither $\bar{C}$ nor $\bar{\mu}$.

Optimality. Recall that a best pair $(\bar{C}^*, \bar{\mu}^*)$ has the smallest cost, $W(\bar{C}^*, \bar{\mu}^*) = W^*$. Two questions about $\mathcal{R}$ remain. Theorem 5 shows that $\mathcal{R}$ contains a best pair, and Theorem 6 compares the cost of the other stopping points with W^*.

Theorem 5

A best pair is a stopping point

Theorem 5 (a best pair is a stopping point)

For any points and any K, there is a best pair $(\bar{C}^*, \bar{\mu}^*)$ in $\mathcal{R}$. In fact, k-means started at the centers of any best pair stops after one learning step, at a best pair.

Proof

Step 1: a best pair exists. There are K^n partitions. For one partition $\bar{C}$, Theorem 1 shows that no centers give a smaller cost than the means of its clusters, and that cost does not depend on the centers of empty clusters. So the smallest cost with the partition $\bar{C}$ is $\sum_{k:\, C_k \neq \{\}} \sum_{i \in C_k} \|x_i - \bar{x}_{C_k}\|^2$. W^* is the smallest of these K^n numbers, and a partition that attains it, with the means of its clusters, is a best pair.

Step 2: in a best pair, every point is with a nearest center. Let $(\bar{C}, \bar{\mu})$ be a best pair, and suppose some point is not in the cluster of one of its nearest centers. Let $\bar{C}^{\text{nc}}$ be a nearest-center partition for $\bar{\mu}$. By Theorem 2, $W(\bar{C}^{\text{nc}}, \bar{\mu}) \leq W(\bar{C}, \bar{\mu})$, with equality only when every point of $\bar{C}$ is with a nearest center, which fails. So $W(\bar{C}^{\text{nc}}, \bar{\mu}) < W^*$, which contradicts that W^* is the smallest cost. So $\bar{C}$ is itself a nearest-center partition for $\bar{\mu}$, and by Theorem 2, $W(\bar{C}, \bar{\mu}) = \sum_i \min_k \|x_i - \mu_k\|^2$.

Step 3: round 1 ends at a best pair. Start k-means at $\bar{\mu}^{(0)} = \bar{\mu}$. Partition step 1 puts every point in the cluster of a nearest center, so by Theorem 2 and Step 2,

$$W\big(\bar{C}^{(1)}, \bar{\mu}\big) = \sum_{i=0}^{n-1} \min_k \|x_i - \mu_k\|^2 = W(\bar{C}, \bar{\mu}) = W^*.$$

By Theorem 3(b), learning step 1 does not raise the cost, so $W(\bar{C}^{(1)}, \bar{\mu}^{(1)}) \leq W^*$. No pair costs less than W^*, so $W(\bar{C}^{(1)}, \bar{\mu}^{(1)}) = W^*$, and $(\bar{C}^{(1)}, \bar{\mu}^{(1)})$ is a best pair.

Step 4: k-means stops there. By Step 2 applied to the best pair $(\bar{C}^{(1)}, \bar{\mu}^{(1)})$, every point is in the cluster of one of its nearest centers in $\bar{\mu}^{(1)}$. So in partition step 2 the tie rule keeps every point where it is, no point moves, and k-means stops and returns $(\bar{C}^{(1)}, \bar{\mu}^{(1)})$. This pair is a best pair in $\mathcal{R}$, and we call it $(\bar{C}^*, \bar{\mu}^*)$.

Implications. By Theorem 5, the smallest cost over $\mathcal{R}$ is W^*: some run of k-means ends at a best pair. The proof finds that run by starting at the centers of a best pair, which are unknown in practice. So Theorem 5 does not give a way to find a best pair. It shows that the stopping rule of k-means does not rule out the best pairs.

Theorem 6

The gap between the stopping points and a best pair

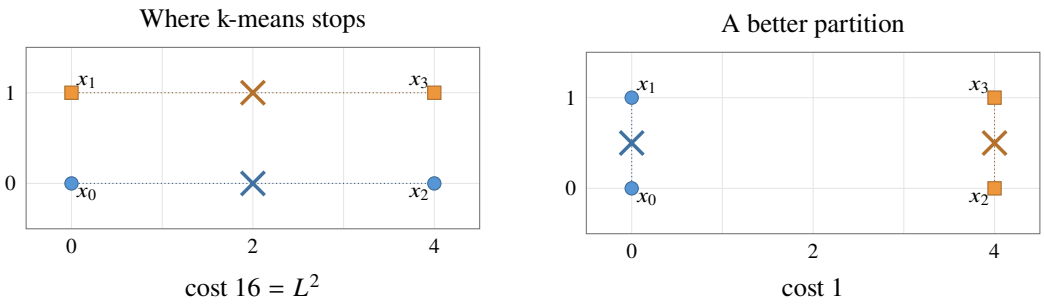
Theorem 6 (no bound on the gap)

(a) Every $(\bar{C}, \bar{\mu}) \in \mathcal{R}$ has $W(\bar{C}, \bar{\mu}) \geq W(\bar{C}^*, \bar{\mu}^*) = W^*$.

(b) For every number $G > 0$, there are four points for which, with $K = 2$, some $(\bar{C}, \bar{\mu}) \in \mathcal{R}$ has $W(\bar{C}, \bar{\mu}) > G \cdot W(\bar{C}^*, \bar{\mu}^*)$.

Proof

Part (a). W^* is the smallest cost over all partitions and all centers, so in particular over the pairs in $\mathcal{R}$.
Part (b). Choose a number $L > 1$ with $L^2 > G$. Take the four points $x_0 = (0, 0)$, $x_1 = (0, 1)$, $x_2 = (L, 0)$ and $x_3 = (L, 1)$, and the starting centers $\mu_1 = x_0$ and $\mu_2 = x_1$.
Step 1: round 1. Point 0 is at squared distance 0 from μ_1 and 1 from μ_2, and point 1 at 1 and 0, so they go to C_1 and C_2. Point 2, at $(L, 0)$, is at L^2 from μ_1 and $L^2 + 1$ from μ_2, so it goes to C_1, and point 3 is at $L^2 + 1$ and L^2, so it goes to C_2. The learning step moves the centers to the means, $\mu_1 = (L/2, 0)$ and $\mu_2 = (L/2, 1)$.
Step 2: round 2. Points 0 and 2 are at squared distance $L^2/4$ from μ_1 and $L^2/4 + 1$ from μ_2, and points 1 and 3 are at $L^2/4 + 1$ and $L^2/4$. No point moves, so k-means stops. It returns $C_1 = \{0, 2\}$, $C_2 = \{1, 3\}$ with $\mu_1 = (L/2, 0)$ and $\mu_2 = (L/2, 1)$, a pair in $\mathcal{R}$ with cost $4 \cdot L^2/4 = L^2$.
Step 3: an upper bound on W^*. The partition $C_1 = \{0, 1\}$, $C_2 = \{2, 3\}$ with the centers $(0, \frac{1}{2})$ and $(L, \frac{1}{2})$ puts every point at squared distance $\frac{1}{4}$ from its center, so it costs 1, and $W^* \leq 1$.
Step 4: W^* is positive. By Theorem 5, a best pair exists. Its two centers can sit on at most two of the four distinct points, so some point is at a positive distance from its center, and $W^* > 0$.
Step 5: the gap. By Steps 2 to 4, $0 < W^* \leq 1$, so the stopping point of Step 2 has cost $L^2 \geq L^2 W^* > G \cdot W^*$.



The proof with $L = 4$. Crosses are the centers, and dotted lines join the points of a cluster.

Implications. Part (b) holds for every G, so no bound of the form $W(\bar{C}, \bar{\mu}) \leq G \cdot W^*$ holds for every stopping point of every input. A stopping point need not even be the best among the partitions that differ from it by one point. In the picture, moving x_1 to the cluster of x_0 and x_2 and recomputing the means lowers the cost from 16 to 34/3. In practice, k-means is run from several starting centers and the run with the smallest cost is kept, and the starting centers can be chosen by the k-means++ rule (see What else is known).

8 The loss is not convex

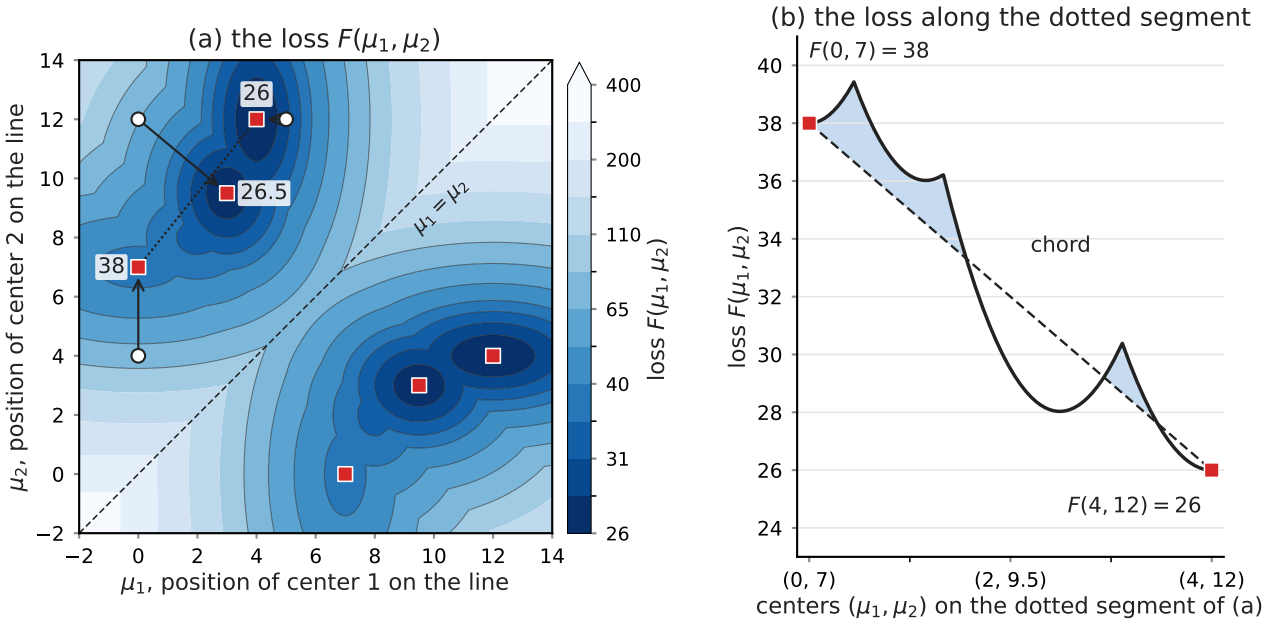
Why k-means can stop above the best cost

We now treat the cost as a function of the centers alone. This function has local minima that are not global minima, which is why k-means can stop above W^*, and so it is not convex.

The loss. By Theorem 2, the best partition for given centers $\bar{\mu}$ is a nearest-center partition. We call its cost the *loss* of the centers:

$$F(\bar{\mu}) := \min_{\bar{C}} W(\bar{C}, \bar{\mu}) = \sum_{i=0}^{n-1} \min_{k=1,\dots,K} \|x_i - \mu_k\|^2.$$

Since W^* is the smallest $W(\bar{C}, \bar{\mu})$ over all $\bar{C}$ and all $\bar{\mu}$, it is also the smallest value of F. When each point has one coordinate and $K = 2$, F depends on two numbers, μ_1 and μ_2, so we can draw it.



The loss for the points at 0, 4, 5, 7 and 12 on a line, with $K = 2$. (a) Red squares are local minima. Arrows lead from three starting centers (circles) to the centers where k-means stops. (b) The loss along the dotted segment of (a). In the shaded parts the curve is above the chord.

Local minima

The dips of the loss, and convex functions

The three dips, each also found with the two centers numbered the other way:

Centers (μ_1, μ_2)	Clusters (points at)	Loss	k-means started at
$(4, 12)$	0, 4, 5, 7 | 12	26, which is W^*	$(5, 12)$
$(3, \frac{19}{2})$	0, 4, 5 | 7, 12	$\frac{53}{2}$	$(0, 12)$
$(0, 7)$	0 | 4, 5, 7, 12	38	$(0, 4)$

With their clusters, all three rows are stopping points in $\mathcal{R}$, and only the first is a best pair.

Where the dips come from. For one fixed partition $\bar{C}$, the cost $W(\bar{C}, \bar{\mu})$ is a bowl with its bottom at the means (Theorem 1). The loss F takes, at each $\bar{\mu}$, the lowest of these bowls, one for each partition, so it has a dip near the bottom of every bowl that is the lowest there. One round of k-means picks the bowl that is lowest at the current centers (partition step) and moves to its bottom (learning step).

The two definitions below apply to any function g that takes a list of centers $\bar{\nu} = (\nu_1, \ldots, \nu_K)$ and returns a number. The loss F is one such function.

Definition 4 (global and local minimum)

Centers $\bar{\mu}$ are a *global minimum* of g if $g(\bar{\mu}) \le g(\bar{\nu})$ for all centers $\bar{\nu}$. They are a *local minimum* of g if there is a number $r > 0$ such that $g(\bar{\mu}) \le g(\bar{\nu})$ for all centers $\bar{\nu}$ with $\|\nu_k - \mu_k\| < r$ for every k.

A global minimum is also a local minimum. In the picture of Section 8, all six red squares are local minima of F, and only the two with loss 26 are global minima.

Convex, informally. For a function of one number, convex means that the straight segment joining two points of the graph, called a chord, never lies below the graph. A convex graph can have flat parts, but it cannot have two separate dips. For lists $\bar{a}$ and $\bar{b}$ of centers and a number λ with $0 \le \lambda \le 1$, we write $(1-\lambda)\bar{a} + \lambda\bar{b}$ for the list with centers $(1-\lambda)a_k + \lambda b_k$. As λ goes from 0 to 1, this list moves along the straight segment from $\bar{a}$ to $\bar{b}$.

Definition 5 (convex function)

The function g is *convex* if, for all lists of centers $\bar{a}$ and $\bar{b}$ and every λ with $0 \le \lambda \le 1$,

$$g\big((1-\lambda)\bar{a} + \lambda\bar{b}\big) \le (1-\lambda)\, g(\bar{a}) + \lambda\, g(\bar{b}).$$

The right side is the chord from $g(\bar{a})$ to $g(\bar{b})$.

Examples with one number. The functions u^2 and $|u|$ are convex. The function $\min\big(u^2, (u-4)^2\big)$ is not: it is 0 at $u = 0$ and at $u = 4$, so the chord is 0 at $u = 2$, while the function is 4 there.

Convexity

For a convex function, every local minimum is global

Theorem 7 (for a convex function, every local minimum is global)

Let g be a convex function of the centers, and let $\bar{\mu}$ be a local minimum of g. Then $\bar{\mu}$ is a global minimum of g.

Proof

Step 1: suppose not. Suppose there are centers $\bar{b}$ with $g(\bar{b}) < g(\bar{\mu})$. Let r be the number of Definition 4 for the local minimum $\bar{\mu}$. Since $g(\bar{b}) \neq g(\bar{\mu})$, $\bar{b} \neq \bar{\mu}$, so the largest distance $D = \max_k \|b_k - \mu_k\|$ is positive.
Step 2: a list close to $\bar{\mu}$. Let $\lambda = \min\left(1, \frac{r}{2D}\right)$, which satisfies $0 < \lambda \leq 1$, and let $\bar{c} = (1-\lambda)\bar{\mu} + \lambda\bar{b}$. For every k, $c_k - \mu_k = \lambda(b_k - \mu_k)$, so $\|c_k - \mu_k\| = \lambda\|b_k - \mu_k\| \leq \frac{r}{2D} D < r$. Since $\bar{\mu}$ is a local minimum, $g(\bar{c}) \geq g(\bar{\mu})$.
Step 3: the chord. By Definition 5, and because $\lambda > 0$ and $g(\bar{b}) < g(\bar{\mu})$,

$$g(\bar{c}) \leq (1-\lambda)\, g(\bar{\mu}) + \lambda\, g(\bar{b}) < (1-\lambda)\, g(\bar{\mu}) + \lambda\, g(\bar{\mu}) = g(\bar{\mu}).$$

This contradicts Step 2, so no such $\bar{b}$ exists, and $\bar{\mu}$ is a global minimum.

Non-convexity

The loss of k-means has a local minimum that is not global

Theorem 8 (the loss is not convex)

There are points for which, with $K = 2$, the loss F has a local minimum that is not a global minimum. For these points, F is not convex.

Proof

Take the points at 0, 4, 5, 7 and 12 on a line, as in the picture. A nearest center by squared distance is a nearest center by distance, so we compare plain distances.

Step 1: $(0, 7)$ is a local minimum. Use $r = \frac{1}{2}$ in Definition 4, and let $\bar{\nu} = (\nu_1, \nu_2)$ with $|\nu_1 - 0| < \frac{1}{2}$ and $|\nu_2 - 7| < \frac{1}{2}$. The point at 0 is within $\frac{1}{2}$ of ν_1 and more than $\frac{13}{2}$ from ν_2. The point at 4 is more than $\frac{7}{2}$ from ν_1 and less than $\frac{7}{2}$ from ν_2. The points at 5, 7 and 12 are less than $\frac{5}{2}$, $\frac{1}{2}$ and $\frac{11}{2}$ from ν_2, and more than $\frac{9}{2}$, $\frac{13}{2}$ and $\frac{23}{2}$ from ν_1. So for every such $\bar{\nu}$, the nearest-center partition $\bar{C}$ puts the point at 0 in C_1 and the other four points in C_2, and $F(\bar{\nu}) = W(\bar{C}, \bar{\nu})$. The means of these clusters are 0 and $(4 + 5 + 7 + 12)/4 = 7$, so by Theorem 1,

$$F(\bar{\nu}) = W(\bar{C}, \bar{\nu}) \geq W\big(\bar{C}, (0, 7)\big) = 0 + 9 + 4 + 0 + 25 = 38 = F(0, 7).$$

Step 2: $(0, 7)$ is not a global minimum. At $(4, 12)$ the nearest centers give $F(4, 12) = 16 + 0 + 1 + 9 + 0 = 26$, which is less than 38.

Step 3: not convex. If F were convex, Theorem 7 would make the local minimum $(0, 7)$ a global minimum, which Step 2 rules out. So F is not convex.

Another way to see it. Numbering the two centers the other way does not change the loss, so $F(12, 4) = F(4, 12) = 26$. The midpoint of $(4, 12)$ and $(12, 4)$ puts both centers at 8, where $F(8, 8) = 64 + 16 + 9 + 1 + 16 = 106$, while the chord there is 26. This check uses only the two numberings of the same best centers. The proof of Theorem 8 shows more: a local minimum with a larger loss, which is the kind of minimum where k-means can stop above W^*.

What this means for k-means. Theorems 5, 6 and 8 fit together. Every run ends at a point of $\mathcal{R}$, some points of $\mathcal{R}$ are best pairs, and others can cost far more. In the picture of Section 8, each stop lies in a dip of the loss, and because the loss is not convex, a dip need not be the deepest. A stop at which some point has two nearest centers need not even be a local minimum. Started at $(4, 5)$, k-means stops at $(2, 8)$ with loss 34, where the point at 5 is at distance 3 from both centers. For $0 < \varepsilon < 1$, $F(2 + \varepsilon, 8) = 34 - 6\varepsilon + 3\varepsilon^2 < 34$, so small moves of the centers lower the loss.

What else is known

Stopping points and optimality beyond these notes

In this section, we collect, without proof, what is known about the questions of Sections 7 and 8.

Question	What is known
How many rounds can k-means need?	Theorem 4 allows K^n. For some sets of points in the plane, with K growing with n and some starting centers, the number of rounds grows exponentially in n (Vattani, 2011). If the points are moved by small random amounts, the expected number of rounds is bounded by a polynomial in n (Arthur et al., 2011), which is closer to what is seen in practice.
Can an efficient method always find W^*?	No such method is known. The k-means problem is NP-hard, already for $K = 2$ when the number of coordinates can grow (Aloise et al., 2009). With one coordinate, dynamic programming finds W^* exactly in time polynomial in n and K, because the best clusters are runs of the sorted points (Wang and Song, 2011).
Can a better start give a guarantee?	Yes, in expectation. Starting centers chosen by the k-means++ rule, which favors points far from the centers already chosen, have expected cost at most $8(\ln K + 2)$ times W^* (Arthur and Vassilvitskii, 2007), and by Theorem 3 the rounds of k-means only lower it.
Is a stopping point at least locally best?	In one sense. A stopping point at which no point has two nearest centers is a local minimum of the loss F, by the argument of Step 1 of Theorem 8, and one with such a tie need not be, as the stop at $(2, 8)$ in Section 8 shows. But moving a single point to another cluster and recomputing the means can still lower the cost, as in the picture of Theorem 6.
Does the best split of a sample approach that of the population?	Yes, under conditions. If the points are drawn at random from a population and the population's best centers are unique, the best centers of the sample approach them as n grows (Pollard, 1981). This is a question of statistics, the subject of the next part of the course.

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